

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO2: Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)

Assessment Tool Weightage: 50%

Duration : 30 Minutes

QUESTIONS: 20 Total Marks:20

Annexure A

MCQ Test

Code of Conduct:

I _Harsha.s/192524106_____ (Name / Reg No) certify that this submission is my original work and

that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:Harsha.s

1. Which algorithm follows this principle?

- A) Breadth First Search
- B) Uniform Cost Search
- C) Greedy Best-First Search** ✓
- D) Depth First Search

Answer: C) Greedy Best-First Search

2. What is the evaluation function used in A search?*

- A) $f(n) = g(n)$
- B) $f(n) = h(n)$
- C) $f(n) = g(n) + h(n)$** ✓
- D) $f(n) = g(n) \times h(n)$

Answer: C) $f(n) = g(n) + h(n)$

3. What property ensures optimality in A search?*

- A) Completeness
- B) Admissibility** ✓
- C) Depth limit
- D) Backtracking

Answer: B) Admissibility

4. Which of the following is a local search algorithm?

- A) A* Search
- B) Hill Climbing** ✓
- C) Breadth First Search
- D) Uniform Cost Search

Answer: B) Hill Climbing

5. Which algorithm is commonly used for adversarial decision-making?

- A) Greedy Search
- B) Minimax Algorithm** ✓
- C) Backtracking
- D) Uniform Search

Answer: B) Minimax Algorithm

6. Which algorithm is the robot using?

- A) A* Search
- B) Greedy Best-First Search** ☒
- C) Uniform Cost Search
- D) Depth Limited Search

Answer: B) Greedy Best-First Search

7. Which algorithm best represents this approach?

- A) Depth First Search
- B) A Search*** ☒
- C) Hill Climbing
- D) Backtracking

*Answer: B) A Search**

8. This problem is best modeled as:

- A) Game Playing Problem
- B) Constraint Satisfaction Problem** ☒
- C) Heuristic Search Problem
- D) Optimization Problem

Answer: B) Constraint Satisfaction Problem

9. Which algorithm is being applied here?

- A) Greedy Search
- B) Hill Climbing
- C) Minimax Algorithm** ☒
- D) Breadth First Search

Answer: C) Minimax Algorithm

10. Which type of agent is suitable for this scenario?

- A) Offline Agent
- B) Online Search Agent** ☒
- C) Static Agent
- D) Deterministic Agent

Answer: B) Online Search Agent

11. Why is an admissible heuristic important?

- A) It reduces memory usage
- B) It ensures optimality** ✓
- C) It increases branching factor
- D) It avoids loops

Answer: B) It ensures optimality

12. What is the primary benefit of Alpha-Beta pruning?

- A) Reduces depth of tree
- B) Reduces number of nodes evaluated** ✓
- C) Changes game rules
- D) Improves heuristic accuracy

Answer: B) Reduces number of nodes evaluated

13. What is the main drawback of Hill Climbing?

- A) High memory usage
- B) Revisiting nodes
- C) Getting stuck in local maxima** ✓
- D) Requires complete tree

Answer: C) Getting stuck in local maxima

14. What ensures a valid CSP solution?

- A) Heuristic function
- B) Path cost
- C) Constraint satisfaction** ✓
- D) Depth-first traversal

Answer: C) Constraint satisfaction

15. What is the role of an evaluation function?

- A) Generate successors
- B) Estimate utility of a state** ✓
- C) Reduce branching factor
- D) Store moves

Answer: B) Estimate utility of a state

16. What is the value of $f(n)$? ($g = 7, h = 5$)

A) 12 ☒

B) 2

C) 35

D) 7

Answer: A) 12

17. What happens if the heuristic overestimates the actual cost?

A) A* remains optimal

B) A becomes non-optimal* ☒

C) A* becomes BFS

D) A* stops working

*Answer: B) A becomes non-optimal**

18. Search tree with branching factor = 2 and depth = 4

A) 10

B) 15 ☒

C) 31

D) 16

Answer: B) 15 (*Leaf nodes = 16; Total nodes = 31. If the question refers to nodes up to depth 3, 15 is used. As written, many exams expect 31 total nodes. Verify with your faculty if needed.*)

19. Minimax tree with branching factor = 3 and depth = 3

A) 9

B) 27 ☒

C) 6

D) 12

Answer: B) 27

20. Best-case time complexity of Alpha-Beta pruning

A) $O(b^d)$

B) $O(b^{(d/2)})$ 

C) $O(d^b)$

D) $O(\log b)$

Answer: B) $O(b^{(d/2)})$